Assignment Instructions

COM3524 Bioinspired Computing

Assignment Title

Modelling a Forest Fire using Cellular Automata

Submission Details

This assignment must be submitted by **5pm on Monday 30th November 2026**. You must submit:

- A recording of your group presentation (maximum 15 minutes).
- A pdf file of your slides.
- A zip file containing your group's adapted code and a readme explaining how to run it.

These items should be submitted via the appropriate named link on Blackboard Ultra. Your total mark will be scaled to represent 50% of the total mark for this module.

Target date for return with feedback

4th January 2027

Formative Assessment

You are requested to submit a single A4 page Work Plan via the Google Form here by **5pm on Monday 9th November 2026** (before the 2nd Forest Fire Computer Lab on Wednesday 11th November). **This plan will NOT receive a formal mark but is an important opportunity for informal feedback prior to completion and submission of your group project.** The informal feedback will be provided during the lab sessions.

A work plan template is provided on Blackboard Ultra. You are encouraged to use the template, but feel free to adapt as you see fit. The work plan should include a list of all the tasks you have identified as being necessary to complete this assignment and a summary of what your collaborative approach will be (i.e. who does what and when).

Task Description

The aim of this assignment is to explore how a Bioinspired Computational Method (Cellular Automata-based Simulation) can be applied to represent and investigate an important real-world system: namely the spread of a forest fire.

You will be working as a group to carry out simulation experiments in order to address the requirements of a fictional client who wishes you to assess the risk of forest fire to their town, and potential ways of mitigating risk. Your starting point for the simulation code will be the 2D

Game of Life (GoL) Cellular Automata simulator used in the lab class in week 5. Before you start, please read the user documentation for the Python-based 1D and 2D Cellular Automata Simulator available on Blackboard Ultra. You need to understand its existing structure, then adapt it to solve the problem described below. You will be working as a group during the remaining lab classes. Note that you may also have to work on this assignment outside timetabled class hours.

Background: Modelling Wild Fires

The propagation of forest fires (wild fires) over particular terrains lends itself naturally to simulation with a Cellular Automata (CA). The array of cells represents the terrain over which the fire can spread, the state of each cell represents whether it is currently on fire (effectively "on" or "off"), and rules of varying complexity can be developed to model the fire spread.

In the simplest implementation, fires are initialised at specific positions on a grid of cells and each cell simply changes state to "on fire" if a neighbour is on fire. Additional complexity can be included by assigning a *fuel resource parameter* to each cell which is depleted over a number of generations, after which the fire goes out. This fuel parameter can be varied over the array of cells to represent different sorts of terrain. Depending on the timescale of the model, cells will either remain "off" indefinitely after the fuel level is depleted (though very long timescale models may include the concept of regrowth).

Additional levels of complexity could allow for any other factors that influence the way that the fire can spread. For instance, the neighbourhood rules can be adapted to introduce the concept of uncertainty by the use of stochastic-based decisions (e.g. the probability of propagation of the fire is based on the number of "on fire" neighbouring cells). In addition, neighbour probabilities can be adapted to model the effect of e.g. gradient and wind direction.

Statement of the problem

A town in the USA is located in a region prone to wildfires. Specifically, town officials have become concerned about the possible fire risk posed to the town by a power generating plant that is situated approximately 50km north-west of the town. A new waste incinerator – another potential source of ignition - has been proposed to be sited approximately 55km north-east of the town. Between the town and these sites there are different types of terrain including:

- Large areas of chaparral - fairly dense, low lying pine vegetation that catch fire quite easily, and each square km can burn for a period of several days.
- large reservoir/lakes
- A canyon containing scrubland or grassland which ignites very easily, each square km burns for a period of several hours
- Dense forest containing trees that don't ignite very easily, but once they are alight, each km^2 can burn for up to one month.

A schematic of the region is shown in Figure 1.

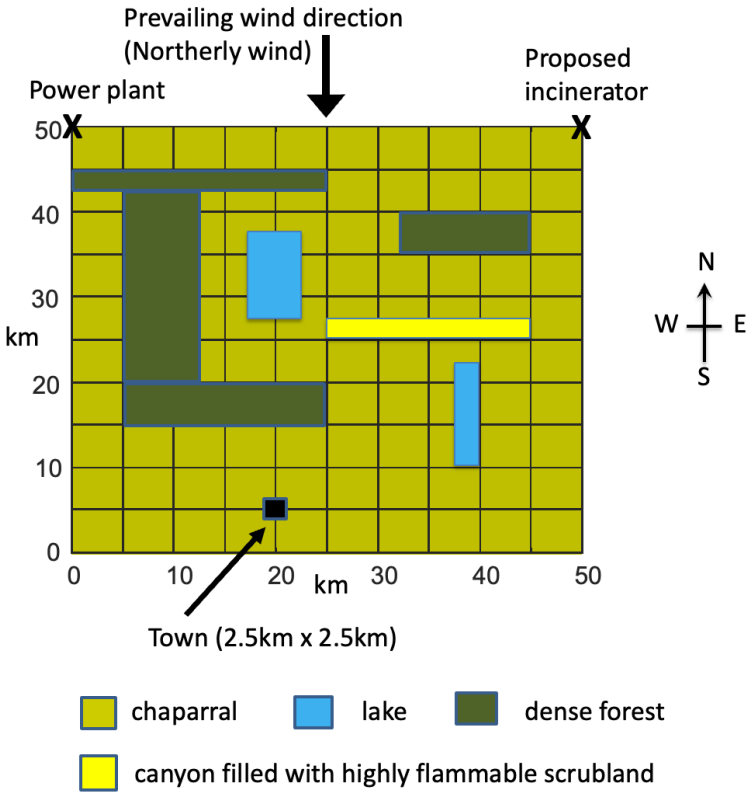


Figure 1: Schematic of region

Client's Requirements

The town officials have commissioned your group of computational environmental engineers to develop a CA-based model to explore potential forest fire spread in this region. In particular, they are interested in exploring the following:

1. The relative time for a fire starting at the power plant or incinerator to reach the town assuming a prevailing wind direction.
2. A study on how time to reach the town from the incinerator would change depending on the wind direction.
3. The planning of short-term intervention in the case of a fire starting at the incinerator. Assume that there is a limited quantity of water (enough to cover only ~12.5km^2) that can be dropped aerially. Use your model to explore where the best place for this drop might be.
4. As a longer-term intervention, it may be possible to increase the existing areas of dense forest. How should it be extended? Are there any other longer-term interventions that you could suggest/model?

In addition, the officials would also like you to conduct a literature review to analyse and summarise the literature on forest file modelling (i.e. present a brief overview of the state of the art of predicting the behaviour of forest fires using computational modelling).

Allocation of Labour

You will need to think about how different parts of each task are allocated between your team members. You should try and ensure that the total workload is evenly distributed. Types of tasks you need to consider will include: analysis of the problem (it makes sense for all group members to do this collectively) – what changes need to be made to the 2D GoL CA code provided to allow you to explore the questions specified?

- searching and analysing and summarising the literature on forest fire modelling
- design and specification of model methodology
- implementation of code modification/development and testing
- specification of model assumptions and parameterisation
- running and producing results of simulations
- Summarising work to address client's questions and critique modelling.

Format

You will be asked to summarise your work in a pre-recorded group presentation, lasting no longer than 15 minutes. A template is provided for the presentation, which you are encouraged to use (you are free to adapt the slide design as you see fit). For the delivery of the presentation, each member of your team should contribute (roughly) equally.

Alongside your group presentation, you will also need to submit a pdf file of your slides and a zip file containing your group's adapted code and a readme explaining how to run it.

Learning outcomes being assessed

The learning outcomes for this module assessed by this assignment are:

- Your ability to apply bio-inspired computing techniques to investigate a real world problem

and

- Your ability to evaluate the accuracy and efficiency of different bio-inspired optimisation approaches to solve a real world problem.

After completing this project, you should be able to work in a group to:

- Analyse a real-world problem and translate this to a list of computational requirements
- Decompose the requirements into a list of computational (and other) tasks
- Implement these tasks effectively by understanding and adapting an existing code-base
- Plan and run computational simulations in order to explore specific questions
- Present your results in a clear and precise way, in accordance with a **15-minute group (recorded) presentation**.
- Communicate your ideas in a way that is clear to a non-computer scientist
- Reflect critically on the limitations of your approach

- Carry out a literature search and summarise work reported in journal papers IN YOUR OWN WORDS (**part of the presentation**)

Note that though some of these skills are specific to this module, many of the above are useful for computer scientists and software engineers in general. You are not expected to have all these skills now but it is expected that you will learn a lot by interaction with the lecturer and demonstrators during the forthcoming lab sessions.

Specific Details

1. This is a team project and you are explicitly expected to demonstrate your ability to work in a team. A good team will work to ensure that all team members have the opportunity to contribute, and have safeguards in place to ensure the project can progress if for some reason a team member is unable to contribute.
2. You should try to ensure that the total workload is evenly distributed among all members of your team. You will need to think about how different parts of each task are allocated between each team member. Here are some examples of tasks you will need to consider: analysis of the problem, literature review, design of the modelling approach, implementation of code modification/development and testing, and running and producing results of simulations.
3. For the delivery of the presentation, each member of your team should contribute (roughly) equally. For example, Background/Literature Review (team member 1); Methods (team member 2); Results (team member 3); Discussion & Conclusion (team member 4).

Top Tips

1. There is no single correct approach to this assignment. You are expected to make your own decisions about parameter values, initial conditions, model implementation, and evaluation methods. However, you should clearly explain and justify your modelling approach and parameter choices. In some cases, reasonable simplifications may be necessary because of simulation time or other practical constraints. If so, you should acknowledge how these simplifications may affect your results.
2. A presentation template is provided as a guide to structure and content. You are encouraged to use the template, but feel free to adapt the slide design as you see fit. Do not exceed the specified time of 15 minutes for your presentation.
3. A good submission should demonstrate achievement across all areas outlined in the marking criteria, which can be found in the document titled *Group_Project_Marking_Criteria.pdf.*
4. Keep on top of the work. This is a significant piece of work which contributes 50% of the final mark. You need to work steadily on it as a team in order to achieve good results.

Referencing and Academic Integrity

The work produced here (including the presentation and code) should be developed by the team and every individual within the team should have a clear understanding of it. As a team, you must ensure that this is the case. If we find any potential academic misconduct, we may conduct interviews and if you are unable to explain to our satisfaction, we will investigate academic misconduct procedures.

Sources of Support

In the first instance, you should seek support in your lab class, and by working together with your team. The module discussion boards on Blackboard Ultra provide a further source of support.

Based on Reading University: Assignment Briefs